Do diseases predict other diseases? A multi-task learning deep neural network approach.

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Abstract

This study is the first to simultaneously use disease prevalence flags as both inputs and targets in a cross-sectional survey setting, applying a masked multi-task learning deep neural network (MTL-DNN) to NHANES 2021-2023 (n=7,269). Conditioning on the disease profiles of 15 chronic conditions alongside 17 non-invasive demographic features, we find that disease prevalence is a meaningful predictor of other diseases, yielding a macro-average AUROC gain of 5 percentage points over a MTL-DNN without disease inputs. The MTL architecture itself provides an additional 3 percentage point improvement over 15 separately trained ANNs, consistent with Caruana (1997) and suggesting the shared trunk learns general comorbidity structure from demographic features alone. Age and self-rated health are the strongest demographic predictors, with cardiovascular comorbidities and the diabetes-kidney disease link representing the most salient disease-to-disease associations recovered. The macro-average AUROC of 0.82 meets or exceeds existing single-disease benchmarks across the majority of conditions, despite using only questionnaire-based inputs rather than laboratory biomarkers or electronic health records.

1. Introduction

The association of disease with other diseases remains inconclusive in the medical literature where most previous studies have focused on investigating the drivers of individual diseases. The interest in uncovering diseases not only lies within the medical field but also in the economic field. As an insurer, if my patient has high cholesterol, how likely is it that they might also suffer from hypertension or diabetes? As a health economist, should we allocate drug X to patient A because they might benefit more than Patient B, as there is a higher probability that they have numerous diseases which benefit from the drug? Thus, understanding the relationships among diseases is vital for efficient allocation (Goldman et al., 2005) .

Beyond understanding such relationships, a model that can capture the associative picture in a single snapshot can be used as a useful screening tool. Given an individual's demographic and their disease profile, how likely is it that they might have another disease? Using a Multi-Task Learning Deep Neural Network (MTL-DNN) allows us to do this well. A significant benefit of using MTL is that NNs can often generalise better (Caruana, 1997) whilst gaining a richer understanding of a particular field. We find this to be the case as the MTL-DNN outperforms all of our individual baseline artificial neural networks (ANN).

Previous literature has broadly focused on single disease prediction, with a notable increase in ANNs entering the space from 2019. A prominent example being the Lopez-Martinez (2020) study which predicts hypertension using an ANN on NHANES data. Over the last few years, studies such as Kim et al (2023) have sought to gain temporal insight by turning to time series NNs, utilising rich electronic health record (EHR) data. A recent and notable paper by Tsai et al. (2025) uses such EHR data (which dates back 10 years) but applies this to the MTL setting. A number of their limitations refer to the reliance on such rich and temporal EHR data. This study will use simple survey and demographic data, such as that in the NHANES setting of Lopez-Martinez (2020) but will apply it to the MTL setting of Tsai et al. (2025). This is applied to the 2021-2023 NHANES data set, predicting 15 prevalent diseases whilst also using prevalence in other diseases to establish disease association among these 15 diseases.

2. Data

The data used is directly collected from the CDC named the National Health and Nutrition Examination Survey (NHANES). The survey is conducted every two years and is intended to be a fair representation of the US population. Our dataset is collated for the 2021-2023 period with 11,933 observations before pre-processing. The data was collected directly from the CDC website and merged on the individual's anonymous ID. 17 variables were selected as demographic inputs, largely dictated by results found in existing literature, such as smoking being harmful, diseases probability increases as we get older, and in emerging literature, how self-perceived health can predict disease outcomes. These non-invasive demographic inputs are used alongside 15 diseases which are used as input features but are also our target variables. The data is cross-sectional and disease diagnoses are self-reported lifetime prevalence, based on questions of the form 'Has a doctor ever told you that you have [condition]?'

3. Model

The main model of interest is a MTL-DNN (Figure 1) where the structure allows us to predict numerous (15) diseases flexibly and learn comorbidity structures more generally. We use two hidden layers to transform the demographic inputs into a shared layer (1), which allows our model to deeply learn general comorbidity patterns from demographic inputs.

$$\mathbf{h} = f_{\text{trunk}}(\mathbf{x}_{\text{demo}}) \tag{1}$$

We then introduce diseases as inputs, ensuring diseases are masked from predicting themselves (pre-diabetes and diabetes are also masked from each other) to prevent the model from relying on this to make predictions, which could be particularly problematic for high prevalence diseases. A third single hidden layer is then introduced which utilises information from the demographics layer and now the diseases inputs (binary) - this makes the masking architecture easier to implement. Finally, we use the aforementioned architecture to predict the disease probability (2) using a standard logistic sigmoid function.

$$\hat{y}_k = \sigma(g_k([\mathbf{h};\, \mathbf{d}_{-k}])) \tag{2}$$

This should strike a balance between learning patterns while maintaining an ability to generalise; to further mitigate overfitting the pre-processed sample of 7,269 is split randomly with proportions 70:15:15, use the validation sample to ascertain when to stop training the model, and we use a dropout rate of 0.3 after the first two activation functions.

Masked binary cross-entropy (3) is used for the loss function which is a natural choice given the binary nature of the prediction problem. Increasing the penalty loss for rare diseases was considered

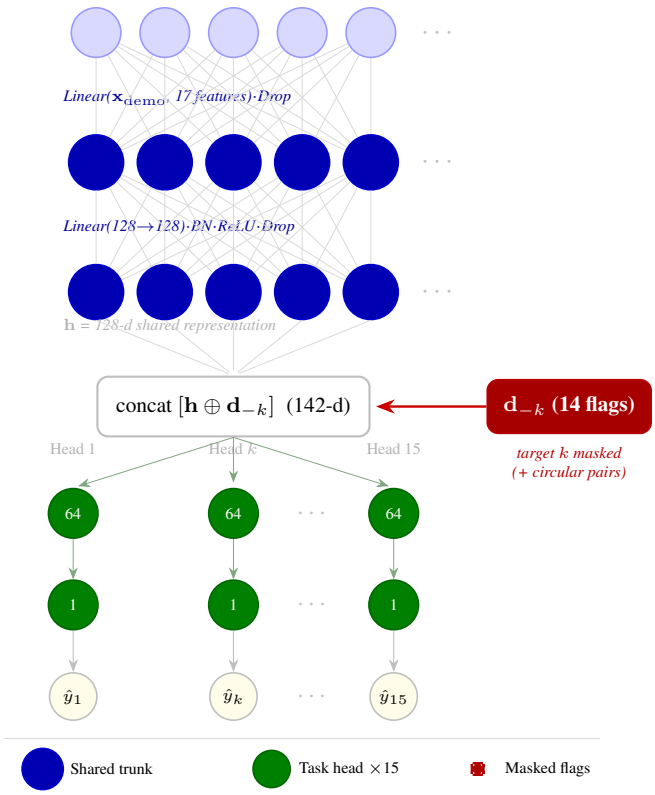

Figure 1: Masked MTL architecture. The shared trunk maps demographic inputs to a 128-d representation $\mathbf{h}$, concatenated with masked disease flags $\mathbf{d}_{-k}$ to form a 142-d input to each task head. Each head predicts $\hat{y}_k \in [0, 1]$ via sigmoid.

as Cheng et al. (2024) suggests MTL models may neglect infrequent diseases. However, we opted against 'cost-sensitive learning' as, in our setting, we found mixed benefits to the PRAUC of infrequent diseases. This may be the result of us already using PRAUC as a validation set hyperparameter, which triggers early stopping, in the training process.

$$\mathcal{L} = \frac{1}{K}\sum_{k=1}^{K} w_k \cdot \left(-\frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} [y_{ik} \log(\hat{p}_{ik}) + (1 - y_{ik}) \log(1 - \hat{p}_{ik})] \right) \quad (3)$$

The results of the model are then compared to a number of baseline models, 15 logistic regressions (LR), 15 random forests (RF), 15 ANNs, and finally a MTL-DNN without diseases as inputs. This allows us to ascertain the general benefits of using a NN in this setting and the individual contributions of using diseases as an input and the MTL architecture.

4. Results

The model meets or exceeds existing benchmark AUROCs across the majority of conditions. This is very well received, not only because it predicts 15 diseases in one model, but also because we generally use less demographic inputs than other studies which can include EHRs, DNA, lab biomarkers, and more than one NHANES data set (use multiple cycles). AUROCs measure sensitivity versus (1-specificity) which means the model did a good job at recalling positive disease cases while also predicting that someone is in fact unlikely to have a disease. Matthews Correlation Coefficients (MCC) also fares well across the board, ranging from 0.21 for stroke prevalence to 0.56 for hypertension prevalence.

Table 1: MTL AUROC on NHANES test set (n=1,091) vs. closest published NHANES binary benchmark.

Disease	Prev.	AUROC	Bench. AUROC	Δ
Diabetes	.103	.883	.830 (11)	+.053
Pre-diabetes	.116	.787	—*	—
Hypertension	.346	.861	.770 (10)	+.091
High cholesterol	.374	.821	—	—
Arthritis	.315	.822	.814 (2)	+.008
Congestive HF	.029	.938	—	—
Coronary HD	.048	.895	.890† (5)	+.005
Heart attack	.042	.840	—*	—
Stroke	.042	.804	.753–.822 (9)	±
Liver condition	.055	.778	—	—
Thyroid problem	.132	.745	—	—
COPD/emphysema	.068	.875	.900† (13)	−.025
Cancer	.141	.754	—	—
Kidney disease	.020	.841	—	—
Asthma	.171	.684	.737 (6)	−.053
Macro avg.	—	.822	—	—

Dashes: no comparable study found albeit may exist. † benchmark includes blood biomarkers.

The baselines comparison reveal a lot about the research question. Foremost, using MTL benefited the model greatly, perhaps due to its learning of general comorbidity patterns (opportunity to learn from more diseases more generally); the model learns the importance of aging and perception of personal health first. On the other hand, we do see that individual LRs, on each disease, using diseases as inputs, often do just as well as MTL-DNN. This may be due to the sample size being too small and thus the model was unable to learn deeper relationships which could be overcome by merging previous NHANES data from 1999. Alternatively, it may be that the relationships in fact weren't that deep, and can be predicted using LRs. To test whether disease-to-disease interactions are non-linear, an additional hidden layer was added to each task head; the improvement in macro-average AUROC was negligible. Therefore, the parity between LR and MTL-DNN suggests that disease co-occurrence in this population may be approximately linear, implying that simple risk scores could replicate the predictive structure without requiring a neural network.

Exploring feature importance gives us a strong indicator which inputs may be associated with a disease. This is done using a simple method where we shuffle inputs values for just that feature and then evaluate the AUROC difference. Alternatives such as SHAP or LIME could have also been used which also have relative benefits and limitations. The two most important inputs are age and perceived personal-health, the latter of which supports the findings of emerging literature (Jylha, 2009) that "self-rated health can reflect the states of the human body and mind". In contrast, categorical values, race and salt intake, held limited importance. Figure 3 includes 4 of the 17 examined demographic inputs and 8 of the 15 disease inputs, where decreases of more than 0.01 in the AUROC are highlighted.

The model does a very good job at recovering what is perceived in the literature. Emphysema is linked to asthma, high cholesterol is linked to diabetes (commonly both led by insulin resistance) and hypertension. Cancer, as an example, is found to have negligible

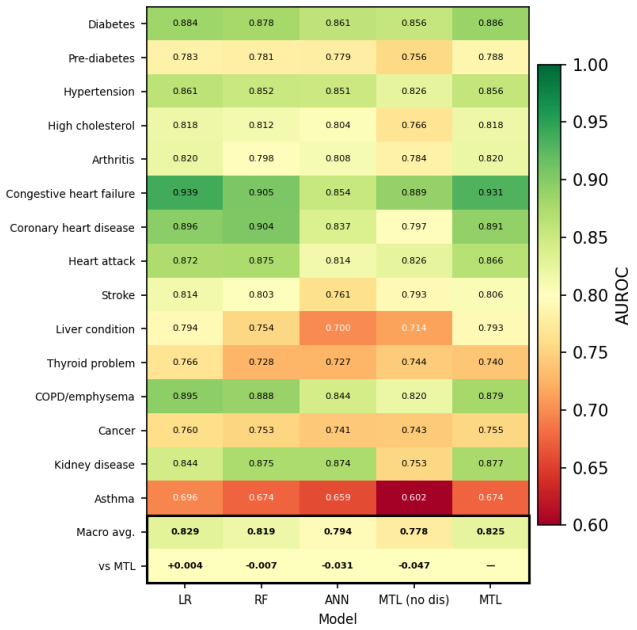

Figure 2: AUROC by model and disease.

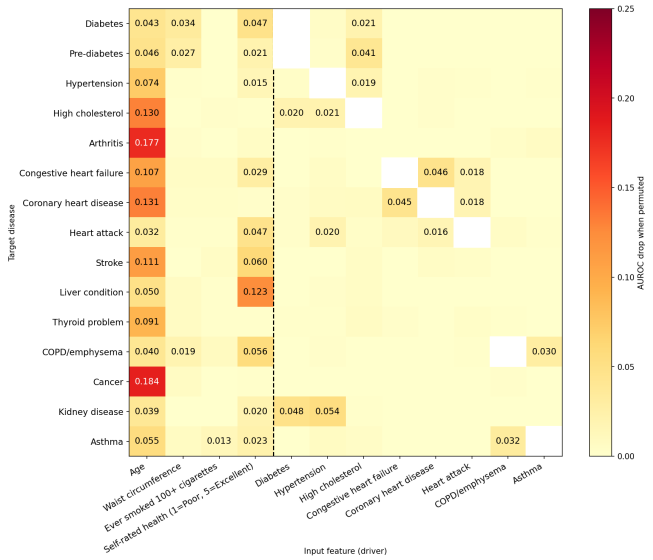

Figure 3: Permutation importance (AUROC drop) for key demographic features and all 15 disease flags.

association to other diseases; perhaps a more granular specifications of prevalence such as 'lung cancer' or 'liver cancer' would reveal relevant associations with the other diseases.

Looking at marginal effects when a disease flag is switched from absent to present can reveal the magnitude of association among diseases, although it is dependent on other relationships within the model, making the interpretability limited. For example, if you have emphysema, holding everything else constant, there is a 10% increase in probability that you also have asthma. Or, if you have one of eight particular diseases, holding everything else constant, there is a substantial (3%-17%) increase in likelihood of high cholesterol prevalence. The latter may be influenced by a lack of dietary variables in the model that drive high cholesterol prevalence, where instead the variation is associated with other diseases, that are also led by dietary choices. On the other side of the spectrum, we find the marginal effect between all diseases and a liver condition to be less than 2%.

5. Discussion

The model does very well given we only use disease prevalence and non-invasive demographic features. However, it is important to recognise previous literature has found great success in using EHRs, DNA, and laboratory biomarkers to improve predictive ability. Whilst EHRs and DNA may be more difficult to source, if available, the model could still benefit from the additional information to improve prediction. There were many cases where we had individuals not responding to specific questions (as they may not know the answer), which would have shrunk our available data down significantly. However, we were still able to retain information from many observations, by NaN masking, when the data was not available. In our study, we refrained from using laboratory biomarkers as they are more invasive, however, it's important to recognise they contain valuable information which we did not benefit from in this study. The aforementioned point translates to laboratory biomarkers too, if an individual were to provide legitimate laboratory biomarkers, the model ought to benefit from the additional information. In both scenarios, it is likely a more sophisticated method than NaN masking would be required, as forward predictions may be biased.

The following limitations merit further consideration. Disease diagnoses are self-reported lifetime prevalence, meaning an individual must have both contracted and received a formal diagnosis to be recorded as positive, introducing systematic under-reporting. Further to this, lifetime prevalence does not capture timing, the model cannot distinguish whether a condition was diagnosed yesterday or 20 years ago, which may obscure the true temporal structure of disease co-occurrence. Finally, whilst NHANES contains hundreds of variables, we deliberately restricted demographic inputs to 17 non-invasive features to avoid circularity and reflection bias, as including variables that are consequences of disease, such as medication use, would risk the model exploiting reverse causation rather than learning genuine predictive structure.

6. Conclusion

Disease prevalence is a meaningful predictor of other diseases, with the MTL-DNN with disease inputs outperforming the equivalent model without them across all 15 conditions. The 5 percentage point macro-average AUROC gain confirms that disease co-occurrence carries predictive signal beyond demographic features alone. The MTL architecture itself contributes a further 3 percentage point gain over separately trained ANNs, consistent with Caruana (1997), suggesting the shared trunk learns general comorbidity structure that individual ANN models cannot recover.

Perhaps the most striking finding is the parity between logistic regression and MTL-DNN when both are given disease inputs. This suggests that disease co-occurrence in this population may be approximately linear, such that simple risk scores could replicate much of the predictive structure without requiring a neural network. This creates a case for using MTL where a researcher opts for a neural network approach, as the unified comorbidity representation learned across all diseases simultaneously may prove more valuable as richer data sources become available.

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